Project Report:
Efficient Text Compression using Huffman Encoding (C++)

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ABSTRACT

This project implements **Huffman Encoding**, a lossless data compression algorithm, using C++ to demonstrate efficient text compression through optimal prefix coding. Huffman Coding is an approach to text compression originally developed by David A. Huffman while he was a Ph.D. student, and published in the 1952 paper "A Method for the Construction of Minimum-Redundancy Codes". It generates variable-length codes. The character with least frequency gets the largest code and vice versa. It utilizes Min Heap and does frequency analysis. The implementation features both encoding and decoding functions, file I/O operations, and statistics reporting including compression ratios. This project shows a real life application of Huffman Encoding in the form of text compression and also exemplify the greedy algorithm. Different data structures like binary trees and priority queues are used.

Introduction

Digital data is growing and requires efficient mechanisms for storage and transmission. Uncompressed text files, especially the ones with repeated use of the same character, are extremely bandwidth- and storage-intensive. This project implements one of the most common lossless data compression algorithms known as Huffman Encoding. Our aim is to provide a practical application of the core data structures and algorithms-the Binary Tree and the Priority Queue-to achieve efficient compression, also keeping in view the data integrity.

Project Objectives

- Implementation of the Huffman Encoding algorithm.
- Frequency analysis and building a sorted frequency table.
- Build Huffman Binary Tree using a Min-Heap Priority Queue.
- To generate Codes by DFS tree traversal.
- Demonstrate encoding and decoding.
- Calculate and report the compression ratio and statistics.

Theoretical Foundation

Huffman coding works by:

- Analyzing and counting character frequencies in the input data
- Building an optimal binary tree (Huffman tree) using a greedy algorithm
- Generating variable-length prefix codes where frequent characters get shorter codes
- Ensuring no code is a prefix of another (prefix property)

Design and Implementation Details

The design and implementation of Huffman Encoding is as follows.

1. Data Structures Used

I. Node Structure

```
struct Node {
    char ch;
    int freq;
    Node* left;
    Node* right;
    Node(char c, int f, Node* l = nullptr, Node* r = nullptr)
        : ch(c), freq(f), left(l), right(r) {}


    ~Node() {
```

```
        delete left;
        delete right;
    }};
```

II. Priority Queue with manual comparator

```
struct Compare {

    bool operator()(Node* a, Node* b) {
        return a->freq > b->freq;    }
};
```

III. Frequency and Code pair

```
struct FreqPair {
    char ch;
    int freq;
};
struct CodePair {
    char ch;
    string code;
};
```

2. File Structure

- Program is divided into files for manageability:
- HuffmanEncoder.cpp (the implementation file)
- HuffmanEncoder.h (header file for the prototypes and data structures definitions)
- main.cpp (main driver program for testing implementation)
- input.txt (for input text)
- compressed_bits.txt (encoded binary string)
- output_decompressed.txt (decoded output text)
- Huffman_report.txt (statistics report file)

3. Core Algorithms Used

I. Frequency Analysis

First the text is read from the file:

```
string readFile(const string& filename) {
    ifstream inputFile(filename);
    if (!inputFile.is_open()) {
```

```
        cerr << "Error: Could not open file " << filename << endl;
        return "";
    }
    string content((istreambuf_iterator<char>(inputFile)), (istreambuf_iterator<char>()));
    return content;

}
```

Then a **frequency count** is analyzed for every character. After counting, a manual sorting algorithm, **Bubble Sort** is applied to sort the unique characters in descending order based on their frequency.

```
void HuffmanEncoder::sortFrequencies() {
    for (int i = 0; i < freqCount - 1; ++i) {
        for (int j = 0; j < freqCount - i - 1; ++j) {
            // Compare adjacent elements and swap if they are in the wrong order
            if (frequencies[j].freq < frequencies[j + 1].freq) {
                // Manual swap block
                FreqPair temp = frequencies[j];
                frequencies[j] = frequencies[j + 1];
                frequencies[j + 1] = temp;
            }
        }
    }

}
```

II. Huffman Tree Construction

a) Initialize a Min-Heap Priority Queue with a node for every unique character.
b) Insert all leaf nodes to priority queue.
c) Until the queue is empty, repeat:

- Extract the two nodes with smallest frequency.
- Create a new internal node using the two smallest nodes as children. And its frequency will be the sum of the two nodes.
- Insert this node back into the Priority Queue.

d) The final node is the root of the Huffman tree.

```
    void HuffmanEncoder::buildHuffmanTree() {
   priority_queue<Node*, vector<Node*>, Compare> minHeap;
    for (int i = 0; i < freqCount; ++i) {
                if (frequencies[i].freq > 0) {

                    minHeap.push(new Node(frequencies[i].ch, frequencies[i].freq));
            }   }
while (minHeap.size() > 1) {
```

```
    Node* left = minHeap.top();
    minHeap.pop();

    Node* right = minHeap.top();
    minHeap.pop();

    int sumFreq = left->freq + right->freq;
    Node* parent = new Node('\0', sumFreq, left, right);
    minHeap.push(parent);

}
```

In the Huffman binary tree structure, the tree is sideways, where the top section is right sub-tree and the root is in middle then the left sub-tree is included.

```
--- Huffman Binary Tree Structure (Sideways) ---
Note: Right branch = '1' path, Left branch = '0' path
                |-- [n, 6]
            |-- (12)
                        |-- [t, 2]
                    |-- (3)
                        |-- [p, 1]
                |-- (6)
                    |-- [o, 3]
        |-- (23)
                |-- [e, 6]
            |-- (11)
                |-- [c, 5]
    |-- (42)
                    |-- [k, 3]
                |-- (5)
                    |-- [m, 2]
            |-- (10)
                |-- [h, 5]
        |-- (19)
                |-- [g, 5]
            |-- (9)
                |-- [i, 4]
(75)
            |-- [Spc, 9]
        |-- (17)
                |-- [a, 4]
            |-- (8)
                |-- [., 4]
    |-- (33)
                        |-- [d, 1]
                    |-- (2)
                        |-- [I, 1]
                |-- (4)
                        |-- [j, 1]
                    |-- (2)
                        |-- [,, 1]
            |-- (8)
                        |-- [w, 1]
                    |-- (2)
                        |-- [y, 1]
                |-- (4)
                    |-- [s, 2]
        |-- (16)
                    |-- [u, 2]
                |-- (4)
                    |-- [f, 2]
            |-- (8)
                |-- [r, 4]
```

III. Code Generation

A **Depth-First Search (DFS)** is applied on the binary tree. While moving to the left child appends '0' to the current code, and if moving to the right child appends '1'.

```
void HuffmanEncoder::generateCodes(Node* curr, string code) {

    if (curr->left == nullptr && curr->right == nullptr) {
        codes[codeCount].ch = curr->ch;
        codes[codeCount].code = code;
        codeCount++;
        return;
    }

    // Recursive Step 1: Traverse left (append '0')
    if (curr->left) {
        generateCodes(curr->left, code + "0");
    }
    // Recursive Step 2: Traverse right (append '1')
    if (curr->right) {
        generateCodes(curr->right, code + "1");
    }}
```

IV. Text Encoding:

When a leaf node is reached, the calculated binary string is assigned as the Huffman Code for that specific character. A binary string of 0 and 1 characters is formed. No code is the prefix of another code.

```
string HuffmanEncoder::getCode(char ch) const {
    for (int i = 0; i < codeCount; ++i) {
        if (codes[i].ch == ch) {
            return codes[i].code;
        }
    }
    return "";
}

string HuffmanEncoder::encode(const string& text) {
    if (text.empty()) return "";
    for (char c : text) {
        updateFrequency(c);
    }
    sortFrequencies();
    buildHuffmanTree();
    generateCodes(root, "");
    string encodedBitString = "";
    for (char c : text) {
        encodedBitString += getCode(c);
```

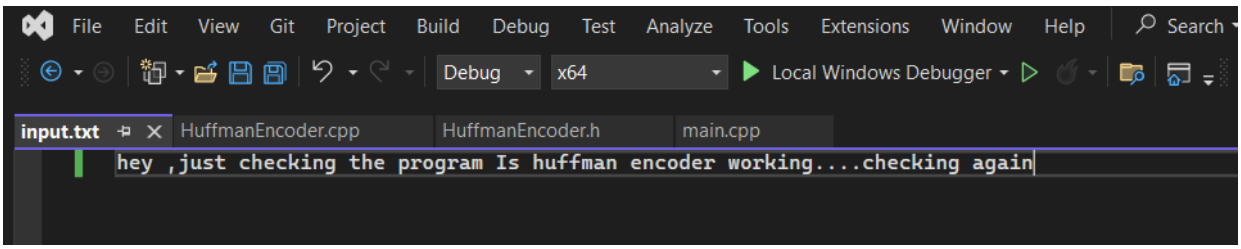
```
    }
    return encodedBitString;

}
```

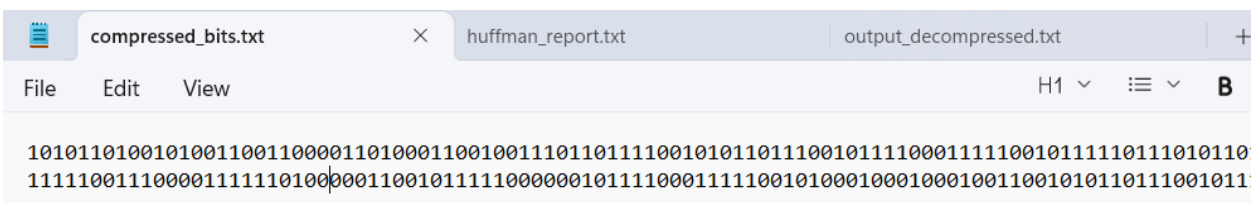
Results and Statistics

File I/O Operations:

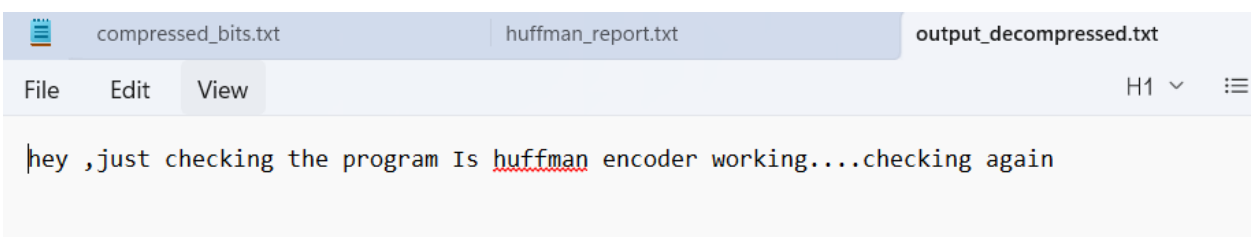
I. Input string in file **input.txt**



II. Encoded string is written to **compressed_bits.txt**



III. Decoded text is written to **output_decompressed.txt**



IV. The program provides detailed compression statistics, shown in the output and written to **huffman_report.txt** file.

compressed_bits.txt | **huffman_report.txt**

File Edit View

```
--- Huffman Binary Tree Structure (Sideways) ---
Note: Right branch = '1' path, Left branch = '0' path
                    |-- [n, 6]
            |-- (12)
                            |-- [t, 2]
                        |-- (3)
                            |-- [p, 1]
                    |-- (6)
                        |-- [o, 3]
        |-- (23)
                    |-- [e, 6]
            |-- (11)
                    |-- [c, 5]
    |-- (42)
                        |-- [k, 3]
                    |-- (5)
                        |-- [m, 2]
            |-- (10)
                    |-- [h, 5]
        |-- (19)
                    |-- [g, 5]
            |-- (9)
                    |-- [i, 4]
(75)
            |-- [Spc, 9]
        |-- (17)
                    |-- [a, 4]
            |-- (8)
                    |-- [., 4]
    |-- (33)
                            |-- [d, 1]
                        |-- (2)
                            |-- [I, 1]
                    |-- (4)
                            |-- [j, 1]
                        |-- (2)
                            |-- [,, 1]
            |-- (8)
                            |-- [w, 1]
```

```
--- Character Frequencies (Sorted) -
Char | ASCII | Frequency
-------------------------------------
 Spc | 32    | 9
  e  | 101   | 6
  n  | 110   | 6
  h  | 104   | 5
  c  | 99    | 5
  g  | 103   | 5
  i  | 105   | 4
  r  | 114   | 4
  a  | 97    | 4
  .  | 46    | 4
  k  | 107   | 3
  o  | 111   | 3
  u  | 117   | 2
  s  | 115   | 2
  t  | 116   | 2
  m  | 109   | 2
  f  | 102   | 2
  y  | 121   | 1
  ,  | 44    | 1
  j  | 106   | 1
  p  | 112   | 1
  I  | 73    | 1
  d  | 100   | 1
  w  | 119   | 1
-------------------------------------
```

```
--- Huffman Codes (The Dictionary) ---
Char | Code
-----------------------------------
  r  | 0000
  f  | 00010
  u  | 00011
  s  | 00100
  y  | 001010
  w  | 001011
  ,  | 001100
  j  | 001101
  I  | 001110
  d  | 001111
  .  | 0100
  a  | 0101
 Spc | 011
  i  | 1000
  g  | 1001
  h  | 1010
  m  | 10110
  k  | 10111
  c  | 1100
  e  | 1101
  o  | 11100
  p  | 111010
  t  | 111011
  n  | 1111
-----------------------------------

--- Compression Statistics (Task 8) ---
Original Character Count: 75 chars
Original File Size (8-bit ASCII): 600 bits
Compressed Bit Stream Length: 323 bits
Estimated Compressed Size (Bytes): 41 bytes
Compression Ratio (Reduction): 46.17 %
---------------------------------
```

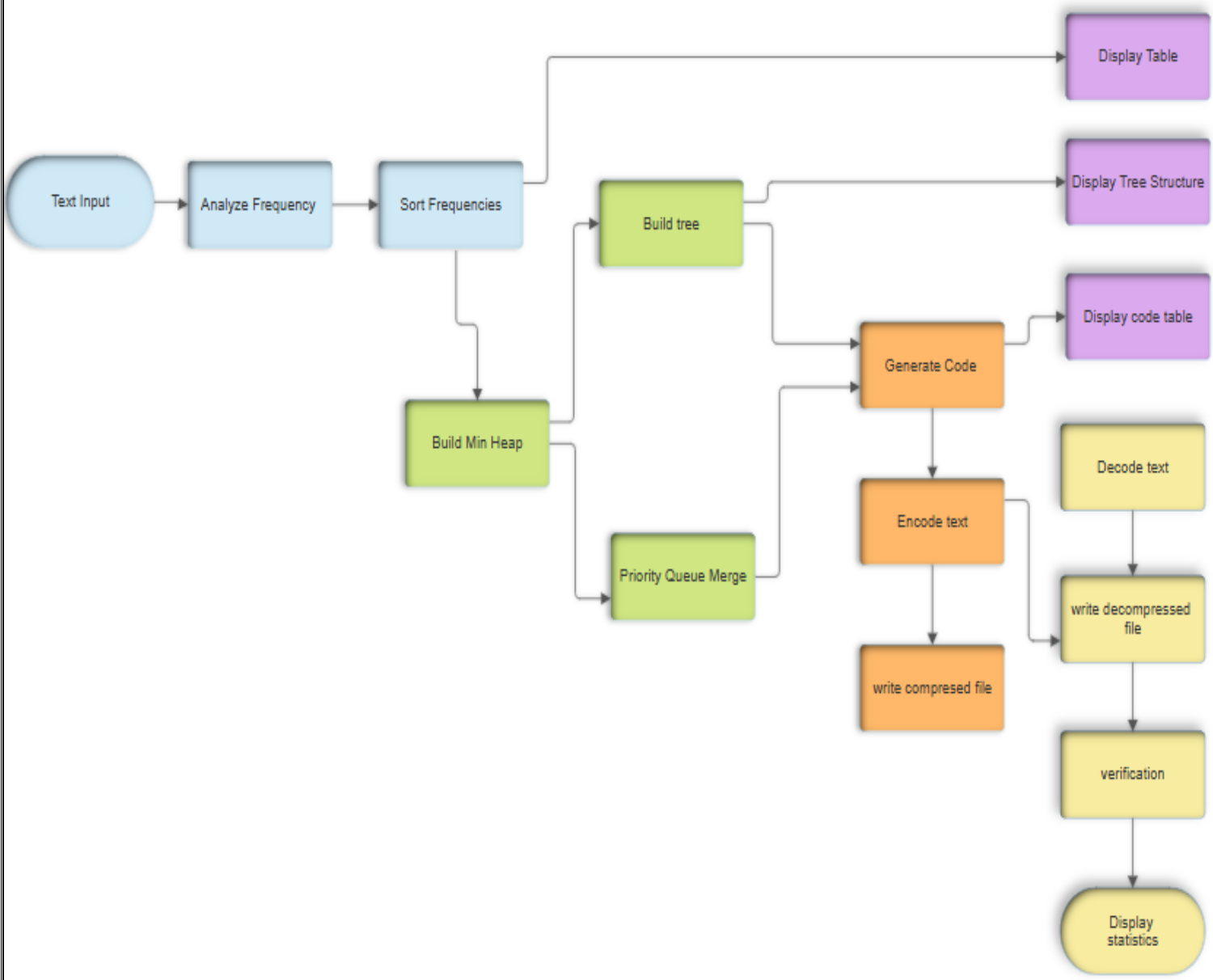
Results:

- **Lossless compression**: Decoded string matches the original string
- **Visualization**: Tree structure display
- **Statistics**: Compression ratio calculation
- **Compression Efficiency:**
  Original File Size (Bits) = Original Length (Characters)* 8 bits/char
  Compression Ratio = (1 - compressed_bits / (original_chars * 8)) * 100%

```
--- Verification and Statistics ---
VERIFICATION SUCCESSFUL: Lossless compression confirmed.

--- Compression Statistics (Task 8) ---
Original Character Count: 75 chars
Original File Size (8-bit ASCII): 600 bits
Compressed Bit Stream Length: 323 bits
Estimated Compressed Size (Bytes): 41 bytes
Compression Ratio (Reduction): 46.17 %
----------------------------------
Full report log written to: huffman_report.txt
```

Flowchart:


Text Input
Analyze Frequency
Sort Frequencies
Display Table
Build tree
Display Tree Structure
Build Min Heap
Generate Code
Display code table
Priority Queue Merge
Encode text
Decode text
write decompressed file
write compresed file
verification
Display statistics

Group Contributions

Core Algorithms & Architecture by Aroona Noor

- Huffman tree construction
- Priority queue implementation
- Special character handling

User Interface & I/O by Muskan Akasha

- Main driver program
- File read/write operations
- Frequency analysis module
- Documentation

Encoding/Decoding & Statistics by Zahra Khan

- Encoding/decoding algorithms
- Compression statistics
- Verification system
- Debugging

Conclusion:

Huffman Encoding is the optimal data compression method which reduces the size of the original data, removes redundant patterns, produce the shortest codes for the most frequent character. In this project the implementation is done with Binary trees and Priority Queue.

CRITICAL QUESTIONS:

Question 1. Is double compression possible?

- What happens if we apply Huffman Encoding again to a Huffman encoded file?
- Would the file become smaller, remain the same, or increase in size?

Answer:

We can apply Huffman again to already encoded text, but after the first Huffman encoding, the bitstream is already close to optimal (with low redundancy).
A second Huffman encoding will not reduce the file size instead its same size or may slightly increase due to overhead from new Huffman encoding.

Question 2. Why is additional compression usually not effective?

- Does the encoded output still contain redundancy?
- How does entropy relate to compression limits?

Answer:
Compression is a process that takes advantage of redundancies like symbols that occur often, recurring patterns, or the use of structure in a particular way.
Using effective Huffman coding, the output now acts like high entropy data, which indicates that each bit is almost impossible to guess.
Entropy is the measure of how much one can theoretically reduce the average number of bits per symbol. Once a data set reaches this limit, it is very difficult to compress it further without removing information.
Thus, Huffman coding leaves the output with almost no redundancy left, making any further compression usually futile.

Question 3. Are there exceptions?

- Are there rare cases where a second compression stage might still reduce size?
- How do multistage compression methods (e.g., ZIP, PNG) work?

Answer:
Although this may not commonly happen, a second compression phase might slightly decrease the size if there are still organized or recurring patterns, like extensive sequences of 0s and 1s, in the data that has been Huffman encoded.
In practice, compression techniques frequently resort to using layered methods, which take advantage of various kinds of redundancy. For instance:

ZIP/Deflate: LZ77 discovers the same sequences first, and afterwards, Huffman coding reduces the symbol frequencies.

PNG: Filtering → LZ77 → Huffman coding. Every single step aims at a specific kind of redundancy.

The application of layered compression techniques is possible due to the accessibility of different algorithms: while one algorithm could be dealing with the compression of patterns, the other could be working on compressing symbol frequencies.

Question 4. Real-world use cases:

Students can explore:

- Layered compression (Huffman + LZ77, Arithmetic Coding, Deflate method)
- Compression pipelines in multimedia formats (JPEG, MP3, PNG)
- Scenarios involving nested or embedded compressed data

Answer:
Layered compression: A lot of modern compress systems put together several algorithms to take advantage of different sorts of redundancy. For example:

- **LZ77 + Huffman (Deflate, ZIP, GZIP):** First, LZ77 finds all the sequences that are being repeated in the data. It shrinks the repeated patterns. Then, Huffman coding takes care of the symbols in the LZ77 output making it even more efficient in terms of bit usage.
- **Arithmetic coding + prediction models:** Formats like JPEG2000 use this method where the statistical models predict the probabilities of the symbols before the actual encoding takes place.

Multimedia formats: The compression pipelines in the audio, image, and video files frequently consist of several stages:

- **JPEG:** The workflow of a JPEG file goes as follows:
  Transform (DCT) → Quantization → Huffman or arithmetic coding.
  Each step, in the transformation, quantization, or coding stage, gets rid of or reduces specific types of redundancy.
- **MP3:** The sound is processed based on human perception **(psychoacoustic modelling) → frequency transform → Huffman coding**.
  Psychoacoustic models filter out sounds that are hardly perceptible to humans and then the remaining symbols are further compressed via Huffman.
- **PNG:** The steps followed are **Filtering → LZ77 → Huffman coding**. The filtering step reorganizes pixel data to show patterns, then LZ77 compresses repeated sequences, and finally, Huffman coding applied to the symbols reduces the redundancies at the symbol level.

Nested or embedded compressed data: Some files have other files that are also compressed taking up space (like a ZIP file containing a PNG or MP3). The multi-stage compression pipelines will still be able to remove remaining redundancy in these embedded data streams and thus achieve an additional size reduction.

Conclusion: Layered compression is efficient because different algorithms identify different types of redundancy which leads overall compression to be more efficient than when a single method is applied multiple times.